

College of Industrial Technology

Seat No.

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1

Year 2025

Subject: 031001123 Chemistry III

Section 15-17

Date: Tuesday, August 5, 2025

Time 10.00-12.00

Examination Setter Name: Assoc. Prof. Dr. Sukanya Thepwatee, Asst. Prof. Dr. Jarukorn Sripradite,
Asst. Prof. Dr. Nisalak Trongsiwat

StudentName.....ID.....Class.....No.....

Directions:

1. This examination paper has not been edited. If any errors are identified, students are advised to exercise discretion in addressing them.
2. The examination is divided into 3 sections. There will be 7 pages (including the cover page), 9 questions and they are worth 90 marks (40 points).
3. Complete all questions in this examination paper.
4. Books, documents and lecture notes are not allowed.
5. Electronic calculators are not allowed.
6. Students must remain in the examination room for no less than 1 hour from the start of the examination.
7. Do not open or start the examination before permission is granted. Students must strictly follow the instructions of the examination.
8. Access to restroom facilities during the examination is not permitted, except in cases of emergency.
9. Students are not allowed to take the examination paper or copy the examination paper out of the examination room. Otherwise, it will be considered cheating.
10. Usage of all communication devices during the examination is strictly prohibited.
11. Smart watches (electronic watches) are not permitted during the exam.

**Cheating in examinations is considered a serious offense and carries the
penalty of expulsion from student status.**

This exam paper is part of a set used to assess student abilities as follows:

		Question (Q)
CLO 1	Explanation and comparison between covalent bond, ionic bond, and metallic bond.	Q2, Q3, Q4, Q5, Q7, Q9
CLO 2	Description of theoretical term and calculation of chemical bond energies.	Q1, Q8
CLO 3	Application of fundamental chemistry in other subjects.	Q6
CLO 4	Realization of discipline, punctuality, and public and self-responsibility.	

Part1 : Total score 30 marks

1. Match the items in Column A with the most correct description or example. One item can be used only once. (5 marks)

A1: Octet rule	A2: Metallic bond	A3: Covalent bond	A4: Ionic bond
A5: Co-ordinate covalent	A6: Exception of octet rule	A7: Lone pair electron	A8: Bonding electron
A9: 16	A10: 18	A11: 20	A12: Resonance

- _____ 1.1) A pair of electrons not involved in bonding
- _____ 1.2) NH_4^+
- _____ 1.3) Total valence electrons of CO_2
- _____ 1.4) The central atom in a molecule has more than eight valence electrons.
- _____ 1.5) Ozone (O_3) is an example of this, where electrons are delocalized.

2. Indicate whether each of the following statements is True (T) or False (F). If false, please correct the following statements. (5 marks)

- 2.1) Sulfur, S, can have an expanded octet due to its position in period 3 of the periodic table.

.....

.....
- 2.2) In Lewis structures, hydrogen atoms can have more than two electrons.

.....

.....

Part 2 : Total score 30 marks

4. Show stable Lewis structures and indicate the molecular geometry using VSEPR theory for the following molecules: (10 marks)

4.1) NH ₃	4.2) PCl ₅
4.3) COCl ₂	4.4) NO ₂

5. Explain why water (H₂O) has a much higher boiling point than methane (CH₄), even though both are small molecules? (5 marks)

.....

.....

.....

.....

.....

.....

.....

.....

วิทยาลัยเทคโนโลยีอุตสาหกรรม

6. Indicate the dominant intermolecular force and rank the following substances in order of increasing boiling point. Explain your reasoning: (10 marks)

Molecules	CH ₄	NH ₃	HCl
Intermolecular force			

Boiling point: _____ > _____ > _____

7. Explain the differences between a metallic bond and an ionic bond. (5 marks)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Part 3 : Total score 30 marks

8. In a combustion reaction, formic acid (HCOOH) is a liquid that reacts with oxygen gas (O₂). It produces carbon dioxide gas (CO₂) and liquid water. Use the following bond energy data (in kJ/mol).

Bond	Energy	Bond	Energy	Bond	Energy
C—C	350	C—O	360	O—H	450
C=C	615	C=O	750	O—O	205
C—H	415	C≡O	1070	O=O	500

8.1) Write the combustion reaction, including the states of all substances, and balance the equation. (2 marks)

.....

.....

8.2) Calculate the value of ΔH for this reaction and identify the type of reaction (endothermic or exothermic). (8 marks)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

มหาวิทยาลัยเทคโนโลยีสุรนารี

8.3) Based on this reaction, if 92 g of formic acid is used, calculate the energy change that occurs. (3 marks)

[illegible]

9. Consider the reaction of Born-Haber cycle for the formation of nickel (II) fluoride (NiF_2):

Given:	First ionization energy of Nickel	=	740	kJ/mol
	Second ionization energy of Nickel	=	1,760	kJ/mol
	Sublimation energy of Nickel	=	400	kJ/mol
	Dissociation energy of F ₂	=	160	kJ/mol
	Lattice energy of Nickel (II) fluoride	=	-650	kJ/mol
	Electron affinity of fluorine	=	-330	kJ/mol

9.1) Write a flow chart of the Born-Haber cycle for the formation of nickel (II) fluoride (NiF_2). (5 marks)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18				
1 H Hydrogen 1.008	Atomic # Symbol Name Weight																2 He Helium 4.0026				
3 Li Lithium 6.94	4 Be Beryllium 9.0122	C Solid Hg Liquid H Gas Rf Unknown																9 F Fluorine 18.998	10 Ne Neon 20.180		
11 Na Sodium 22.990	12 Mg Magnesium 24.305	Metals Nonmetals																15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798				
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29				
55 Cs Caesium 132.91	56 Ba Barium 137.33	57-71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)				
87 Fr Francium (223)	88 Ra Radium (226)	89-103	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)				
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																					
57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97							
89 Ac Actinium (227)	90 Th Thorium 231.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium 237.04	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)							